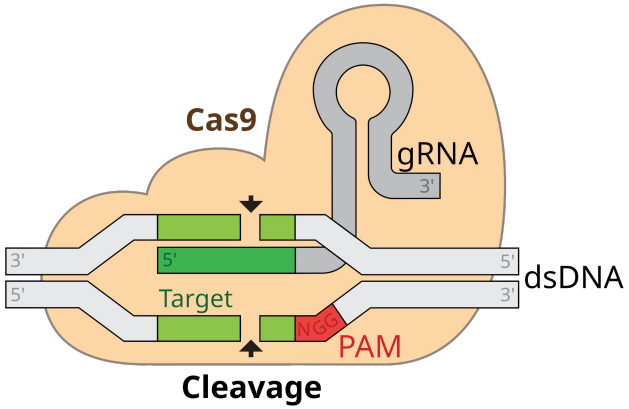
BME 105: Genetics in the Genomics Era

Lecture 19: Gene editing



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Lecture 19 Learning Objectives

- Understand how antibiotic and toxin selection systems work
- Understand <u>how</u> and <u>why</u> conditional knockouts are created
- Be able to describe the CRISPR-Cas9 mechanism of gene editing
- Learn how to design the crRNA region of guide RNA to target a gene of interest
- Understand the implications of how accurate and precise CRISPR-Cas9 is, compared to earlier methods
- Be able to modify the genetic engineering principles you've learned to make knockins containing a specific sequence with:
    - LoxP/Cre recombination after antibiotic selection
    - CRISPR-Cas9 targeted editing

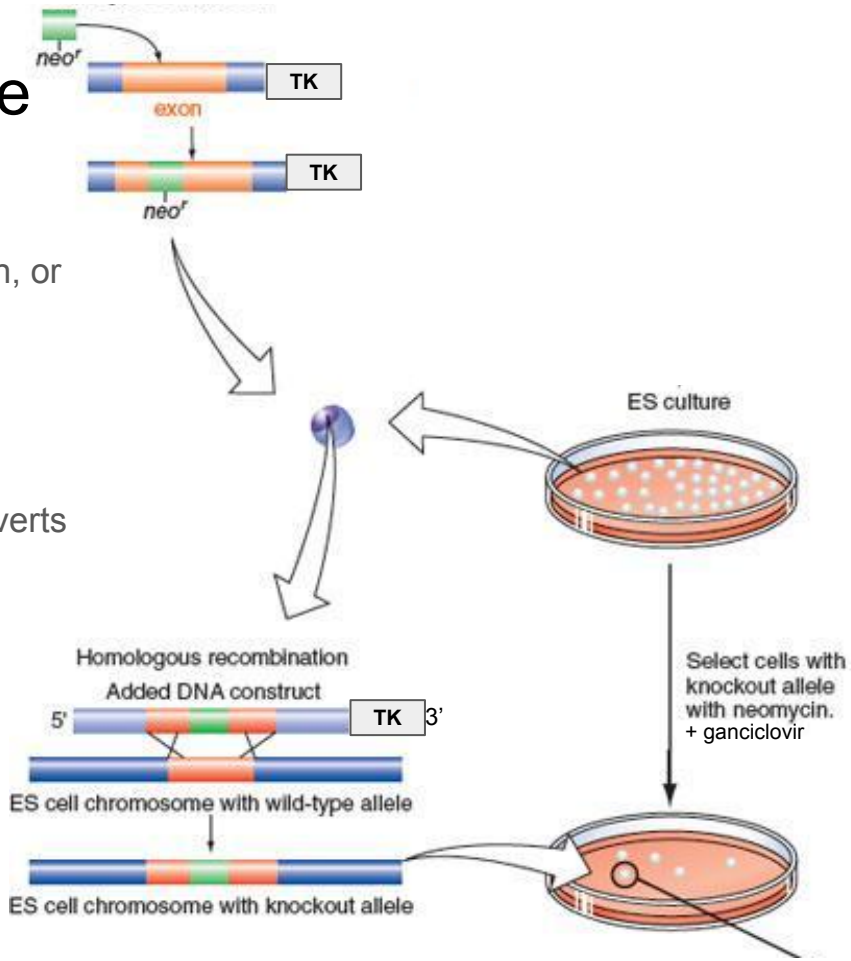
Review: Making a knockout mouse

Problem: need to clear cells with incorrect genotype
Agouti ES cells without the knockout either have no insertion, or *neo* is inserted in the wrong place

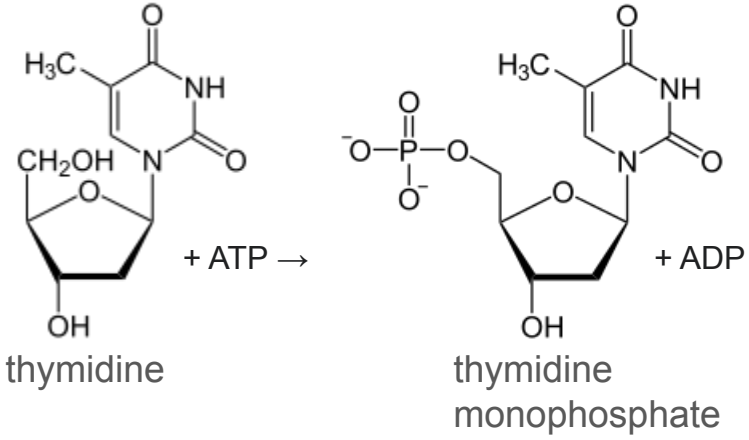
Solution: poison cells with incorrect insertion
Neomycin resistance + TK gene

Herpes simplex virus thymidine kinase (HSV-TK) converts the prodrug ganciclovir (GCV) to a toxic metabolite

Homologous recombination is very inefficient



The TK suicide gene system



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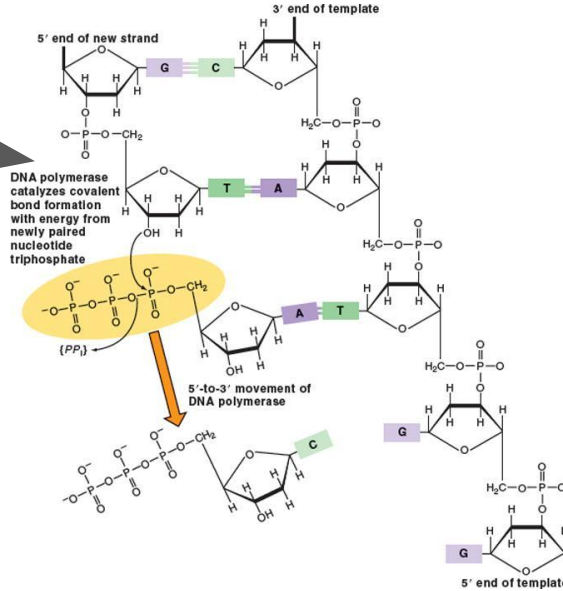
Neomycin resistance + TK gene

Herpes simplex virus thymidine kinase (HSV-TK) converts the prodrug ganciclovir (GCV) to a toxic metabolite

TK ⇒ The enzyme responsible for initiating the phosphorylation of deoxythymidine to deoxythymidine triphosphate for incorporation into nascent DNA.

HSV-TK: needed by the virus for reactivation from ganglionic neurons during the latent stage of its life-cycle, since these cells express only very low levels of endogenous mammalian tk.



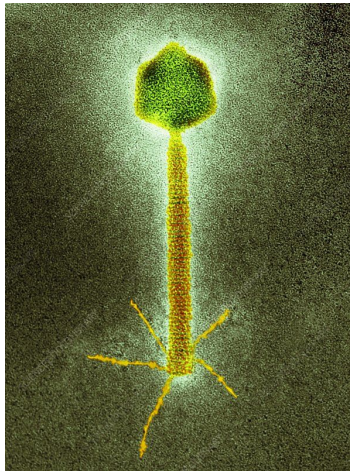
Making a conditional knockout mouse

What if you wanted to be able to control when and where the target gene was knocked down?

- Gene knockout might be lethal
- Gene knockout might have pleiotropic effects, so you want to study the knockout in a specific tissue or condition
- Knockin allows you to see the effect of a mutation in the endogenous context

Making a <u>conditional</u> knockout mouse

P1 prophage
- Non-integrative
- DNA genome maintained in host as circular plasmid



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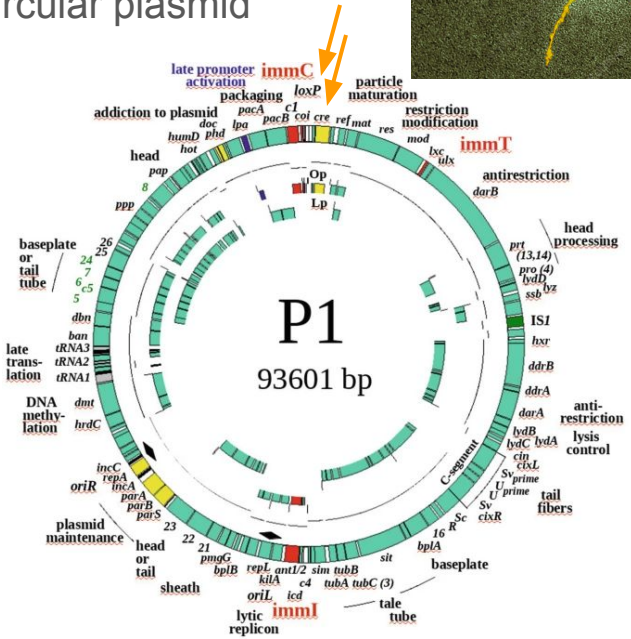
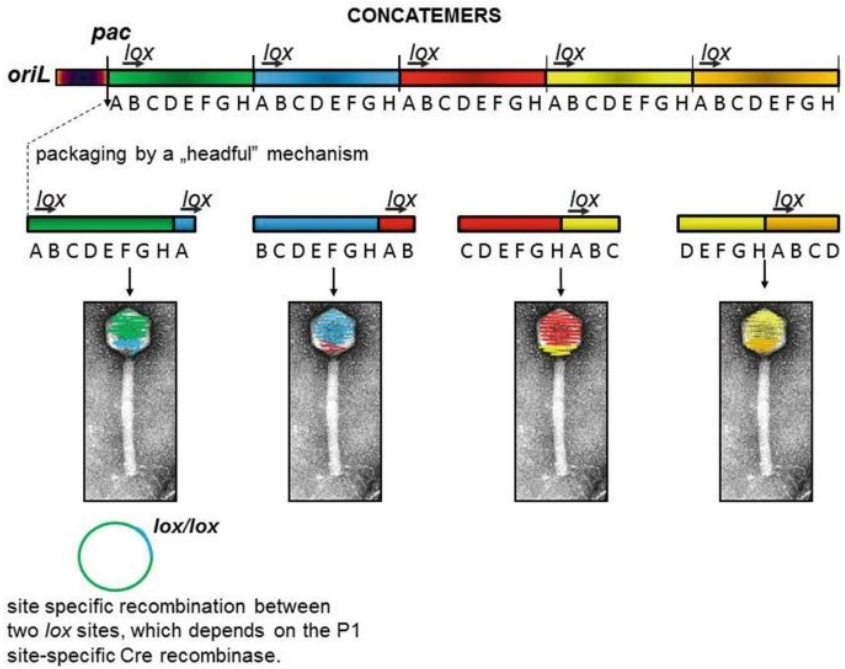


A(nother) transgenic system for controlling transgene expression

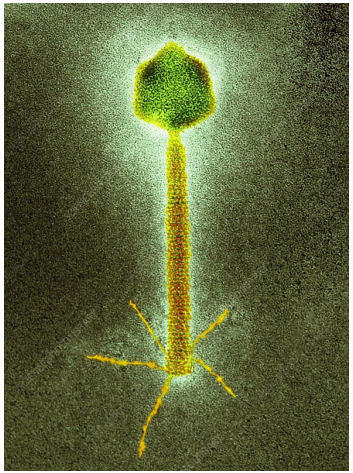
Cre-lox system: Two components derived from the P1 bacteriophage
- Cre recombinase
- LoxP recombination sites

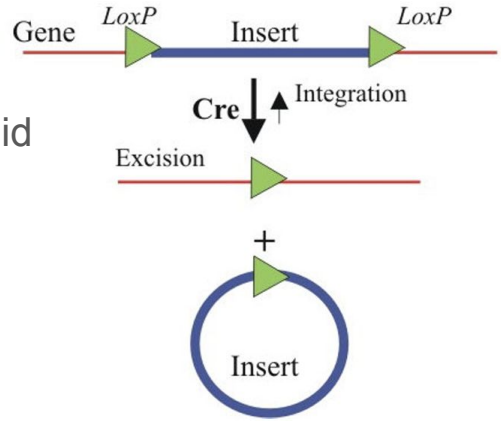
Making a <u>conditional</u> knockout mouse



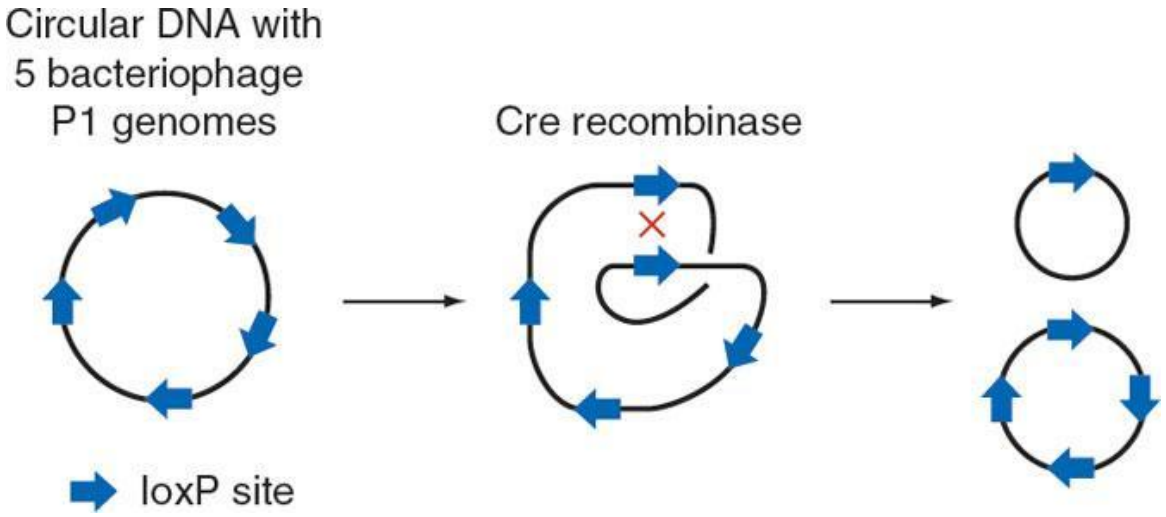
P1 prophage

- Non-integrative
- DNA genome maintained in host as circular plasmid
- Linear P1 genome is packaged in multiple units in the capsid
- *Lox*-specific Cre recombination circularizes plasmid

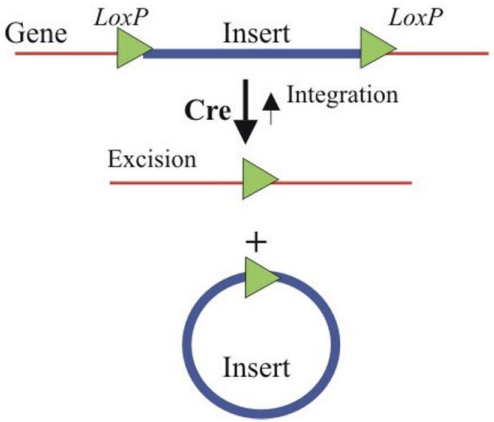




Using the Cre/loxP system for conditional knockout mice



Sequences flanked by LoxP sites are described with the term “floxed



Conditional knockout mice with Cre-Lox recombination

LoxP site

Add LoxP sites on either side of (i.e., straddling) neo and gene of interest

TK=thymidine kinase

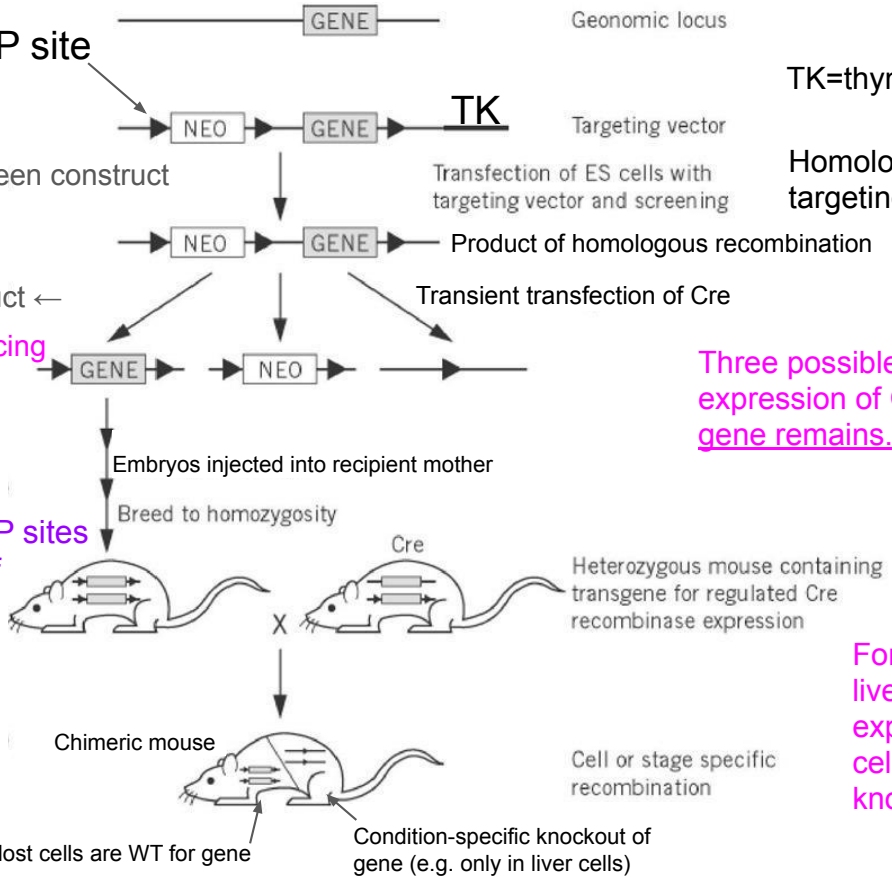
Rare homologous recombination between construct and nuclear genome inserts transgene

Homologous recombination to introduce targeting vector in endogenous locus

neo allows selection of insertion ←

No *tk*, no cell toxic ganciclovir byproduct ←

Transfect Cre-expressing vector, inducing recombination between LoxP sites



Three possible products after transient expression of Cre. <u>Select for case where the gene remains.</u> Identify through DNA analysis.

Wild type with loxP sites straddling gene of interest

For example: If Cre is under a liver-specific promoter, mouse expresses Cre only in liver cells, and so the gene is only knocked out in the liver

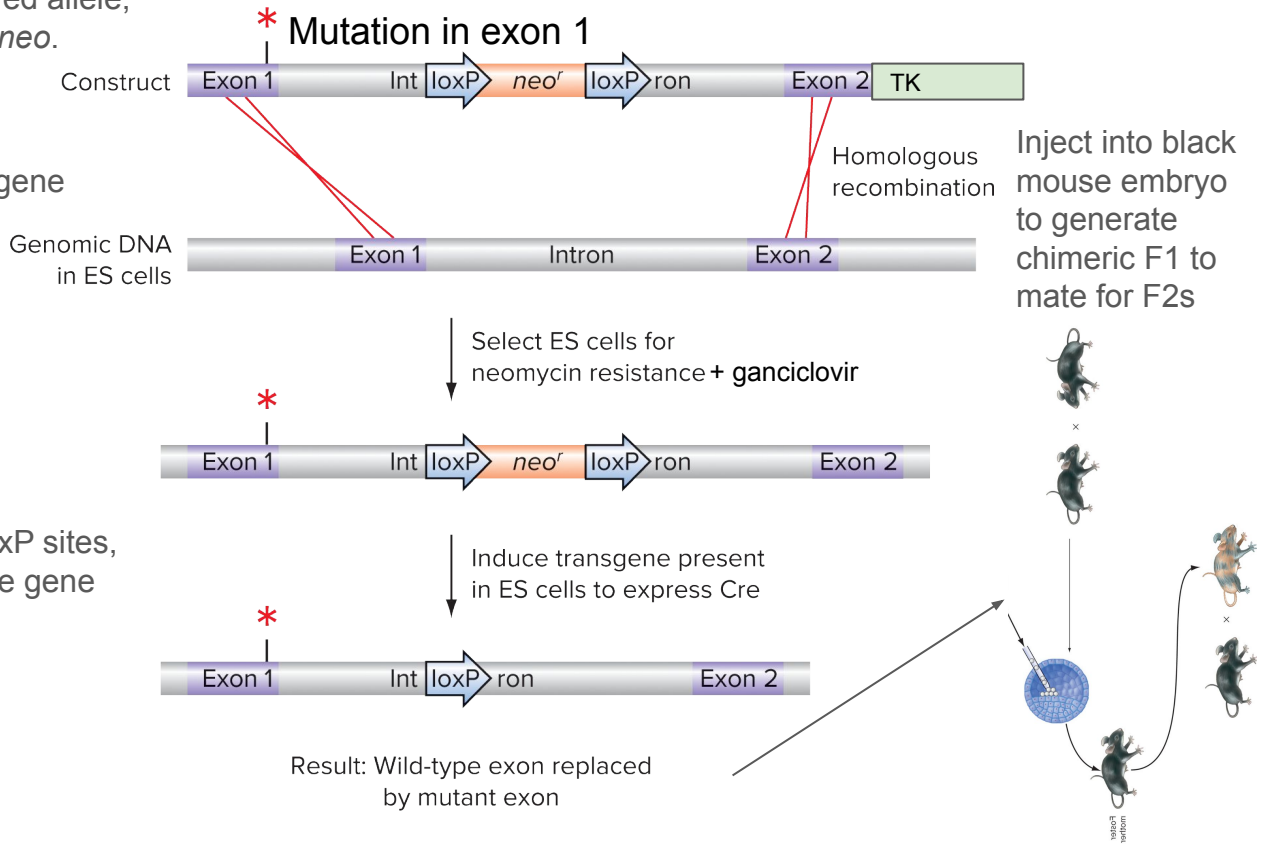
Knockins to make specific mutations

Construct: homologous sites straddling altered allele, intron sequences, and loxP sites straddling *neo*.

Rare homologous recombination between construct and nuclear genome inserts transgene

neo allows selection of insertion ←
No *tk*, no cell toxic ganciclovir byproduct ←

Induction of Cre expression to recombine loxP sites, removing neo, leaving the new version of the gene



Knockins to make specific mutations

Rare homologous recombination between construct and nuclear genome inserts transgene

Can we make this step more efficient?

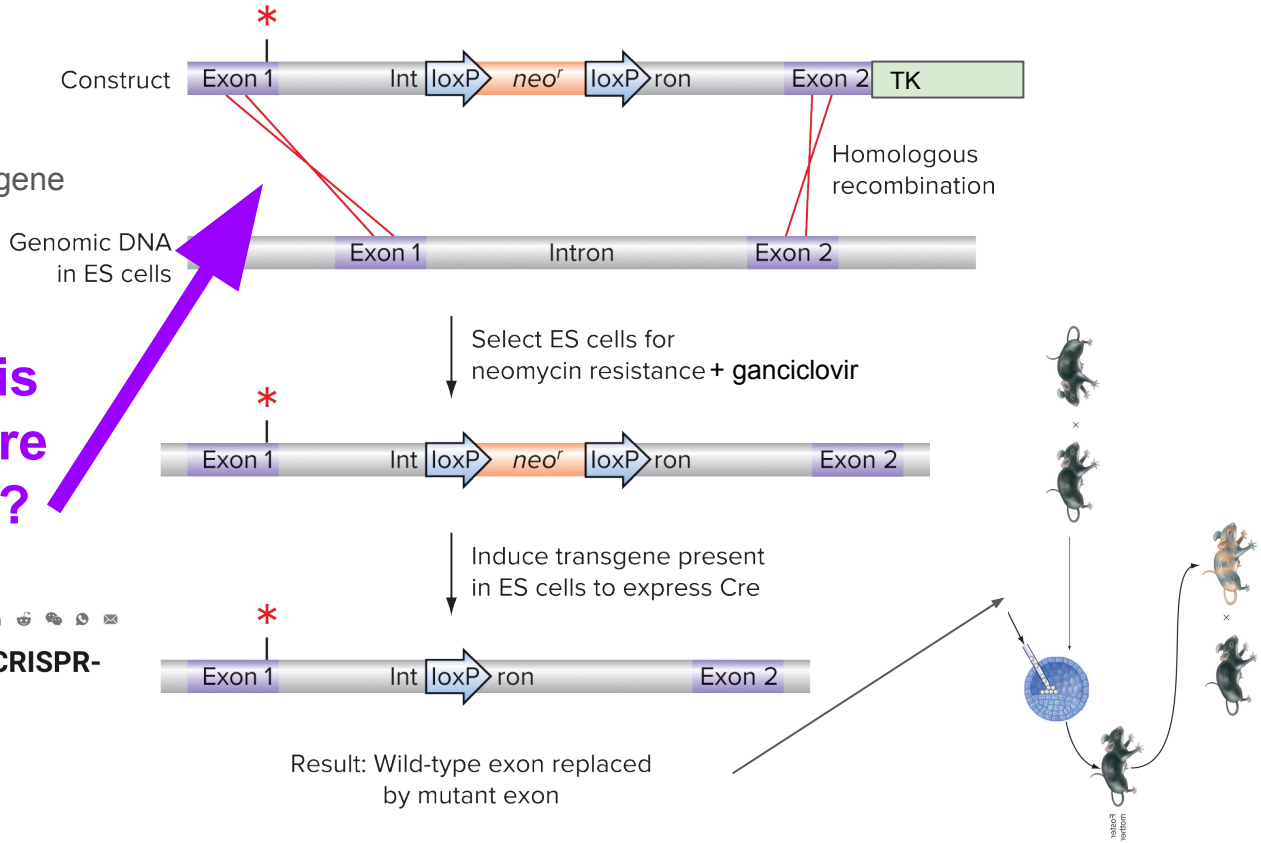
Yes! As of ~2014, with CRISPR-Cas9

REVIEW

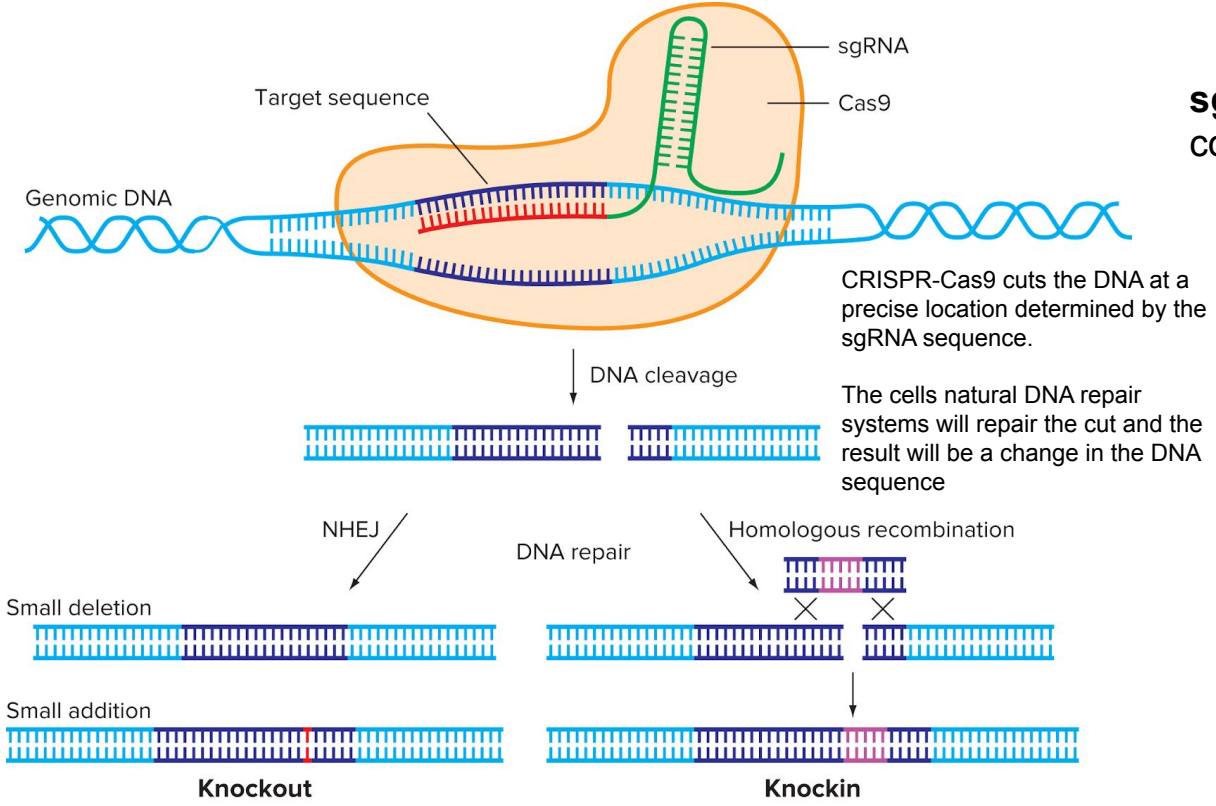
The new frontier of genome engineering with CRISPR-Cas9

JENNIFER A. DOUDNA AND EMMANUELLE CHARPENTIER Authors Info & Affiliations

SCIENCE • 28 Nov 2014 • Vol 346, Issue 6213 • DOI: 10.1126/science.1258096



More efficient genome editing with CRISPR-Cas9



sgRNA is composed of:

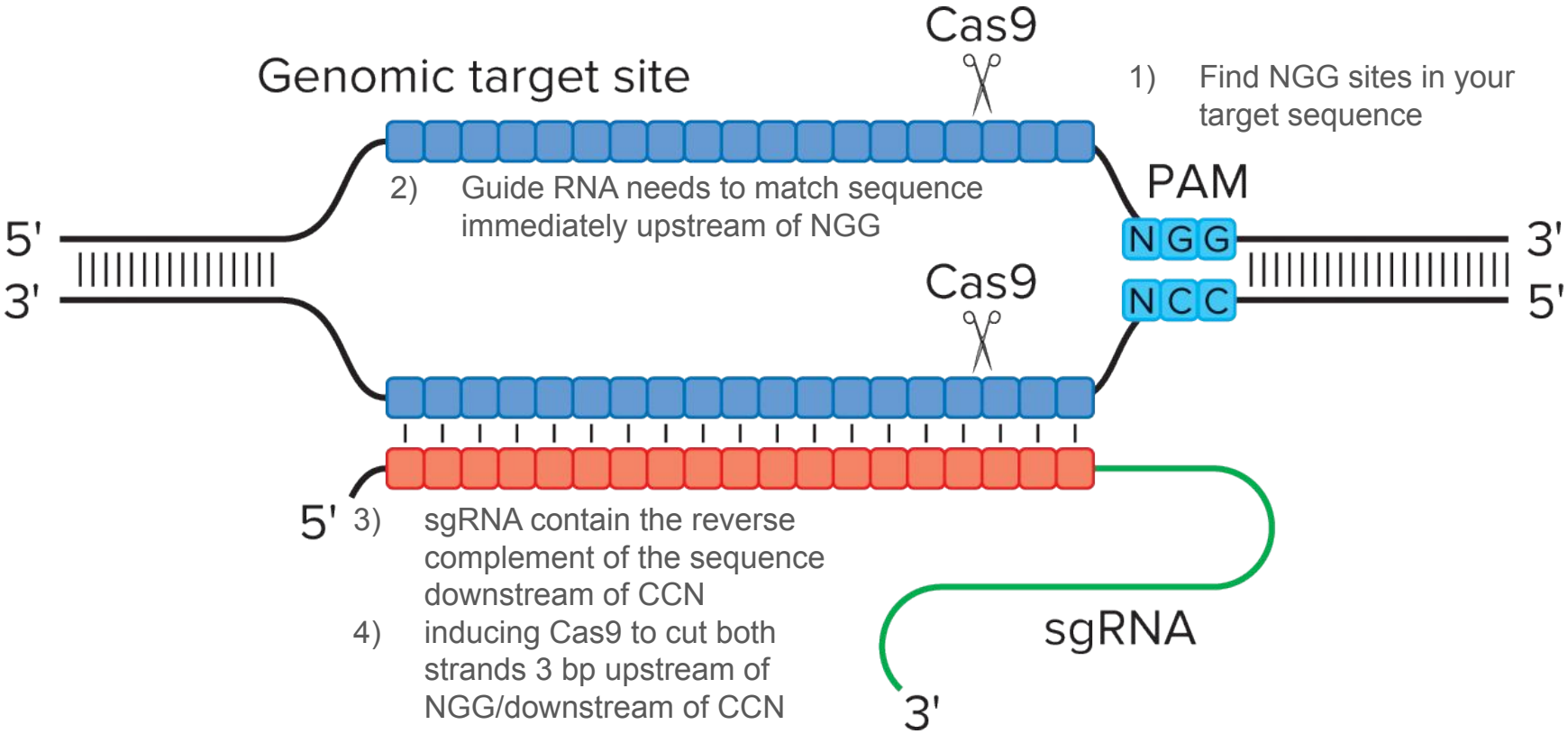
crRNA - portion that base-pairs with DNA

tracrRNA - important for association with Cas9 protein

Non-homologous end joining (NHEJ) repair will result in many different sequences.

⇒ Different cells will have different edits.

Target site in genome needs a PAM sequence (NGG)



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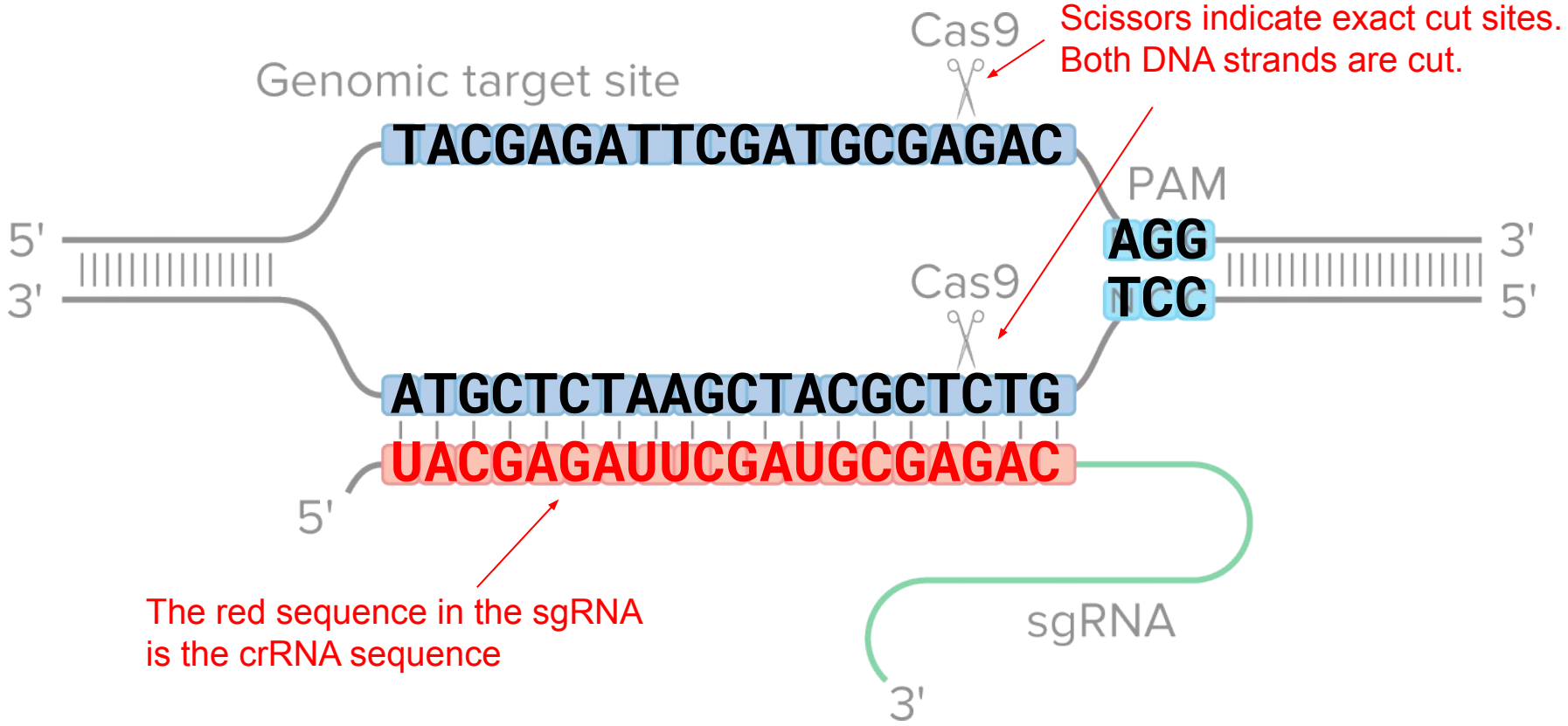
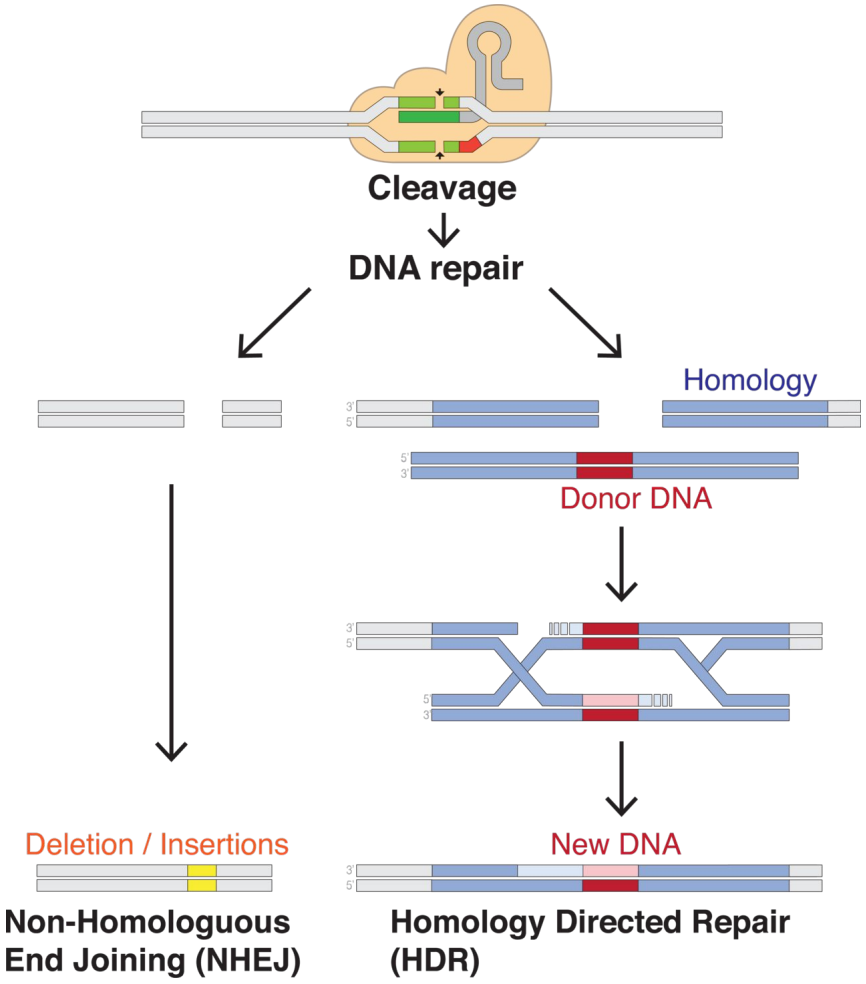
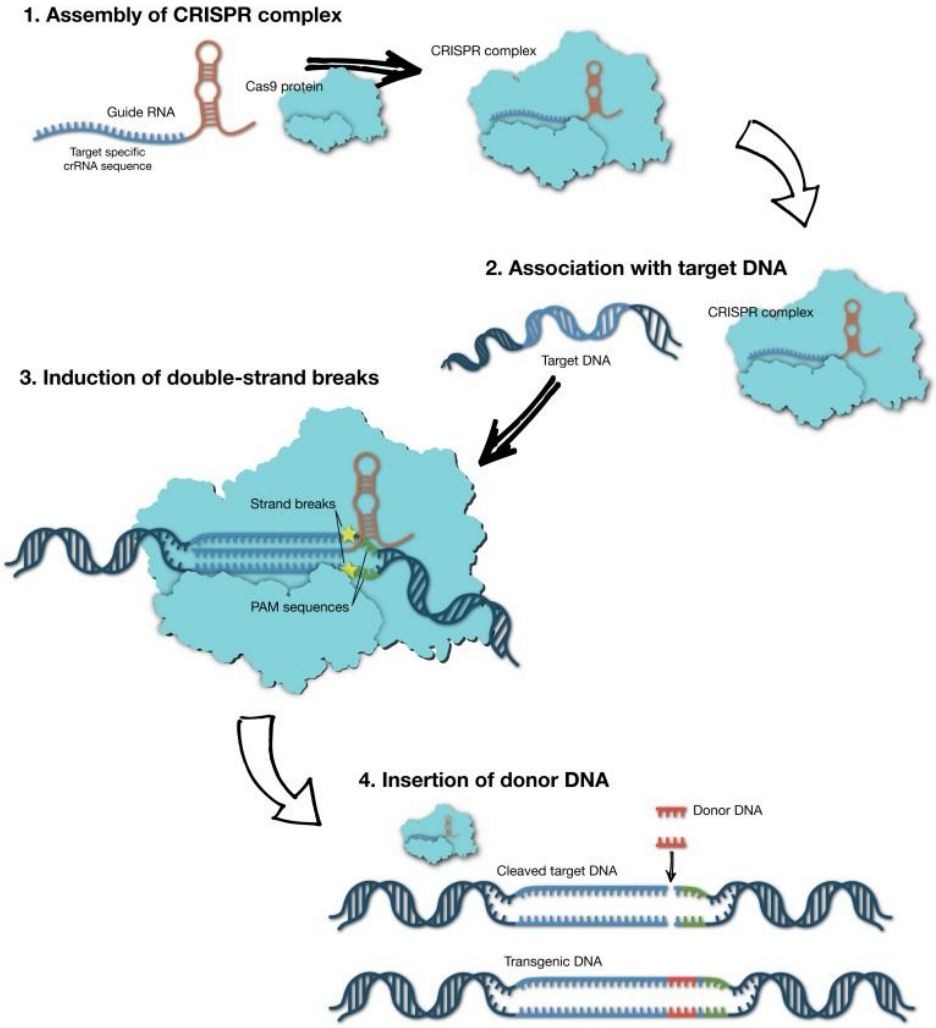


Two repair outcomes produce two edit types: indels or replacements

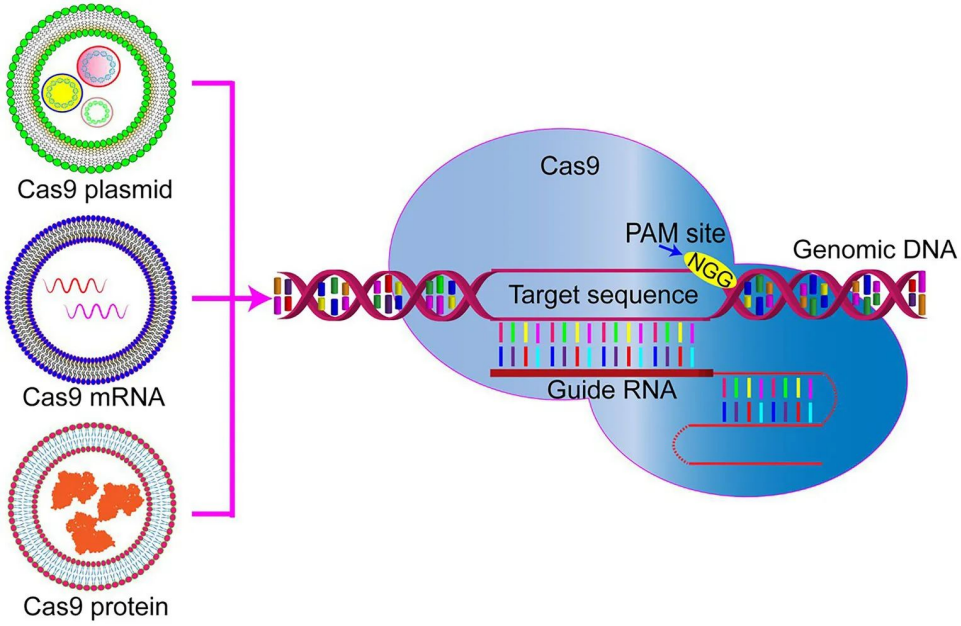


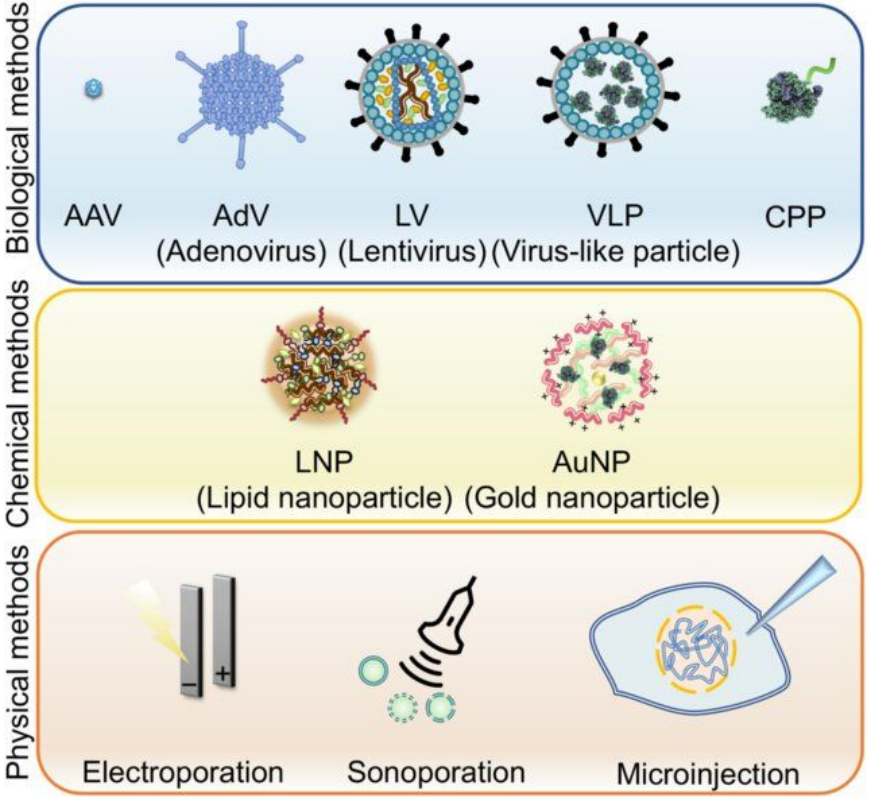
The steps of CRISPR-Cas9 targeted gene editing



Delivery of CRISPR-Cas9 components

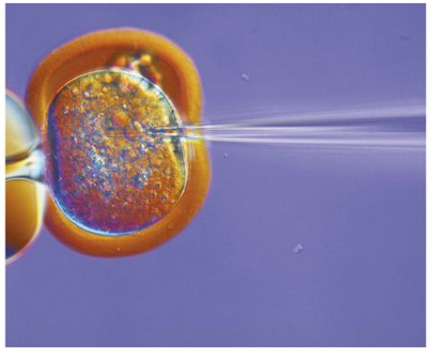
Cas9 can be delivered as DNA, mRNA, or protein
sgRNA sequence can be delivered as DNA or RNA





CRISPR makes it more efficient to make a knockout (KO) and knockin(KI) mice

(a) Injecting a transgene into a recently fertilized mouse egg



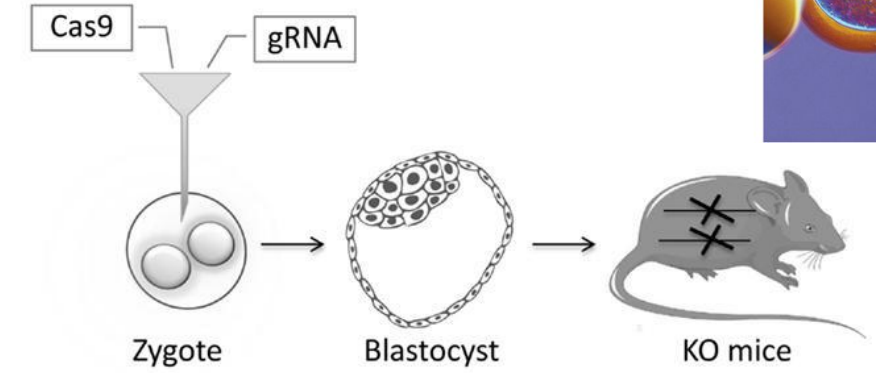
No need for ES cells!

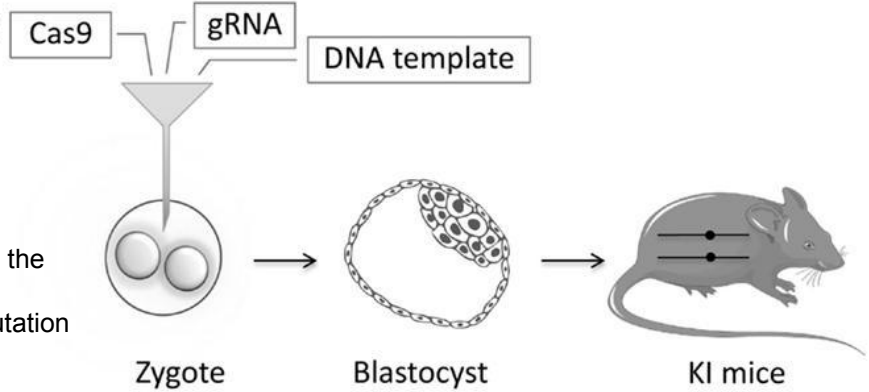
Components can be added as transgenes for pronuclear injection

Zygote develops into blastocyst.

The researcher will need to sequence or genotype the F1 mouse to confirm edit of interest.*

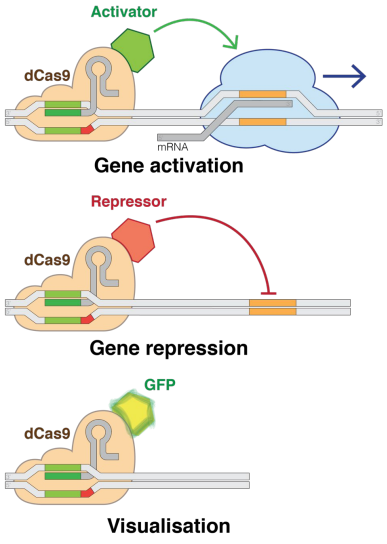
May need to perform series of crosses to make homozygous KO and KI mice.

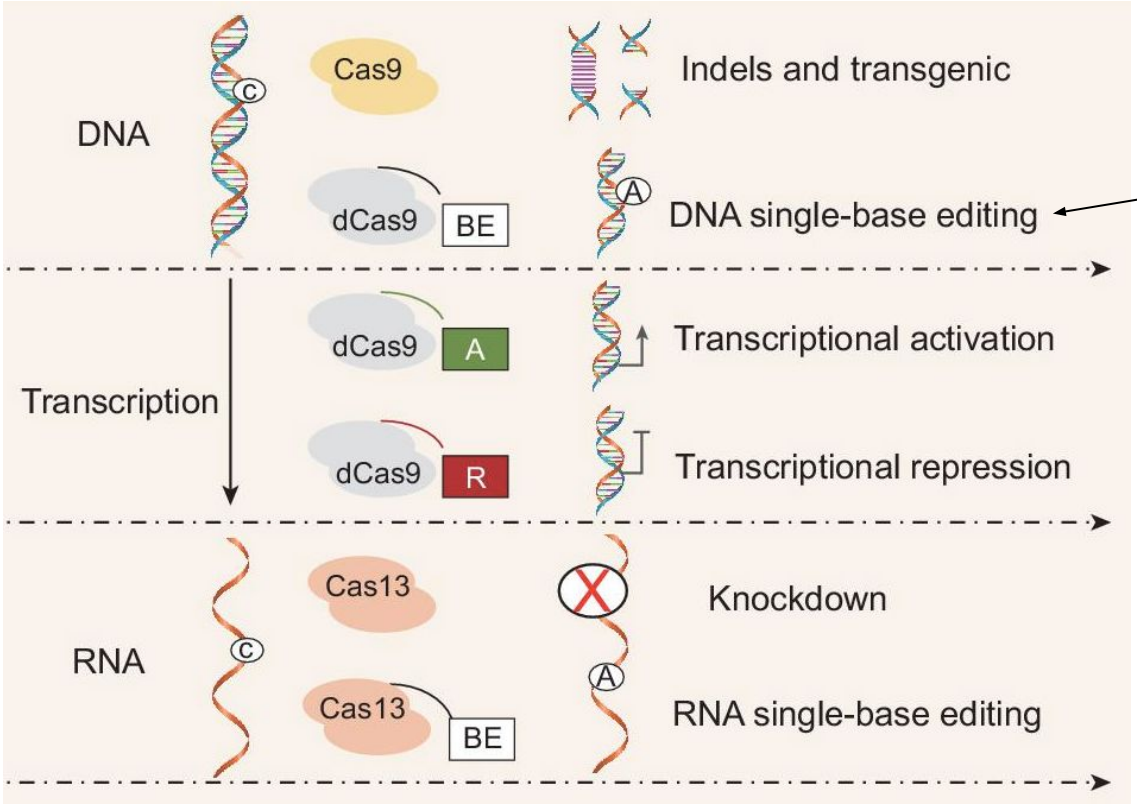




*Pronuclear editing does not result in a chimera; however, the result may be (i) different mutations on both copies of the chromosome (ii) the same mutation on both copies (iii) mutation on one copy of the chromosome, or (iv) no mutation at all.

There are many many more CRISPR/Cas9 systems!





System used to treat KJ Muldoon, the baby with the genetic disorder

SHOTS - HEALTH NEWS

A promising genetic treatment tailor-made for a baby born with a rare disorder

MAY 15, 2025 - 1:02 PM ET

HEARD ON ALL THINGS CONSIDERED

Rob Stein

5-Minute Listen + PLAYLIST TRANSCRIPT

KJ Muldoon, who was born at Children's Hospital of Philadelphia in August, reaches out to doctors after being treated for a rare genetic disorder using CRISPR technology.

CHLOE DAWSON/Children's Hospital of Philadelphia

CRISPR-Cas9 evolved as a bacterial immune system for recognizing and cleaving phages they had encountered before

REPORTS

CRISPR Provides Acquired Resistance Against Viruses in Prokaryotes

RODOLPHE BARRANGOU, CHRISTOPHE FREMAUX, HÉLÈNE DEVEAU, MELISSA RICHARDS, PATRICK BOYAVAL, SYLVAIN MOINEAU, DENNIS A. ROMERO, AND PHILIPPE HORVATH Authors Info & Affiliations

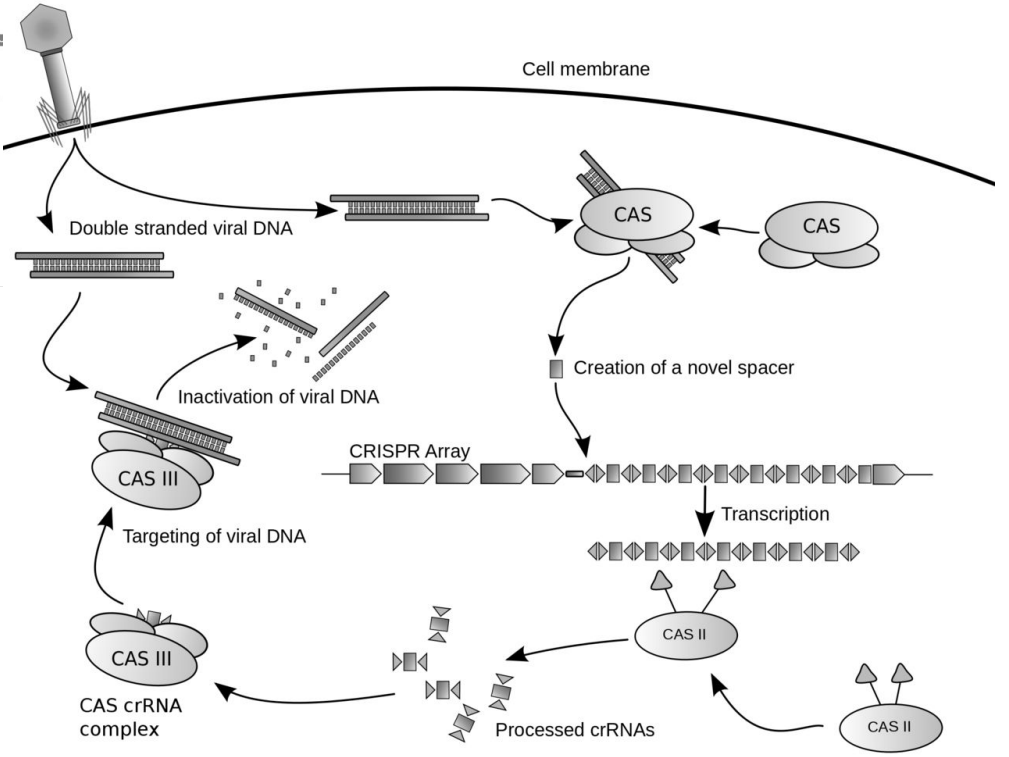
SCIENCE · 23 Mar 2007 · Vol 315, Issue 5819 · pp. 1709-1712 · DOI: 10.1126/science.1138140

REVIEW

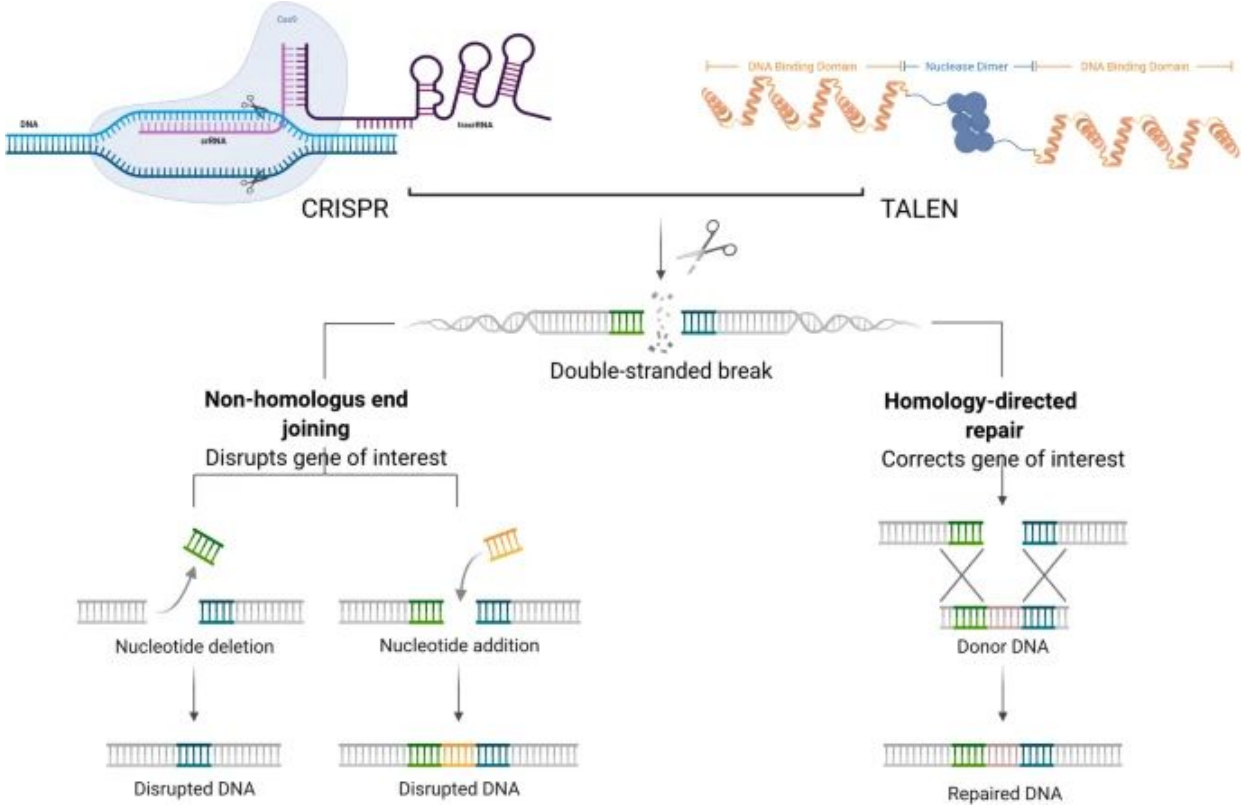
CRISPR/Cas, the Immune System of Bacteria and Archaea

PHILIPPE HORVATH AND RODOLPHE BARRANGOU Authors Info & Affiliations

SCIENCE · 8 Jan 2010 · Vol 327, Issue 5962 · pp. 167-170 · DOI: 10.1126/science.1179555



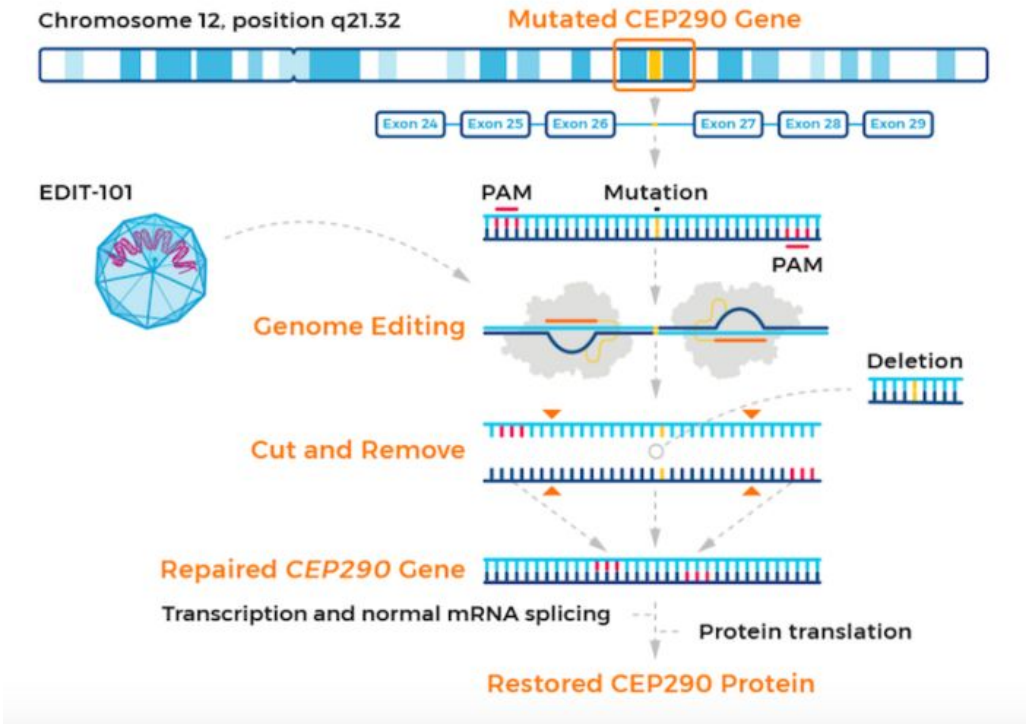
Downstream of dsDNA cleavage, CRISPR and TALEN are similar (leverage cellular DNA repair mechanisms)

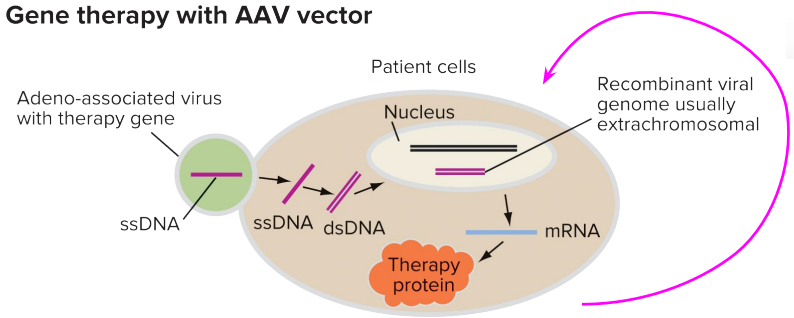


Preliminary gene editing experiments restore vision

Leber congenital amaurosis 10 (LCA 10):

- The emerging treatment targets a specific mutation, c.2991+1655A>G in Intron 26, of the gene CEP290.
- Removing the mutation is sufficient to restore normal splicing and gene function





Expression of sgRNA and Cas9 to remove LCA10 allele in CEP290